

Technology in the Home

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Home Technology Shopping List

Shopping

For new consumer device:

High Performance
Low Power
Security

Models
Ecosystem

Rapid Integration
Low Risk

- Next-generation home devices must deliver increasing convergence without compromising user experience and cost expectations
- How can the industry meet these requirements while containing risk and development time scales?

ARM Processor Roadmap

- ARM offers a roadmap of processors that meet future performance requirements

Shopping

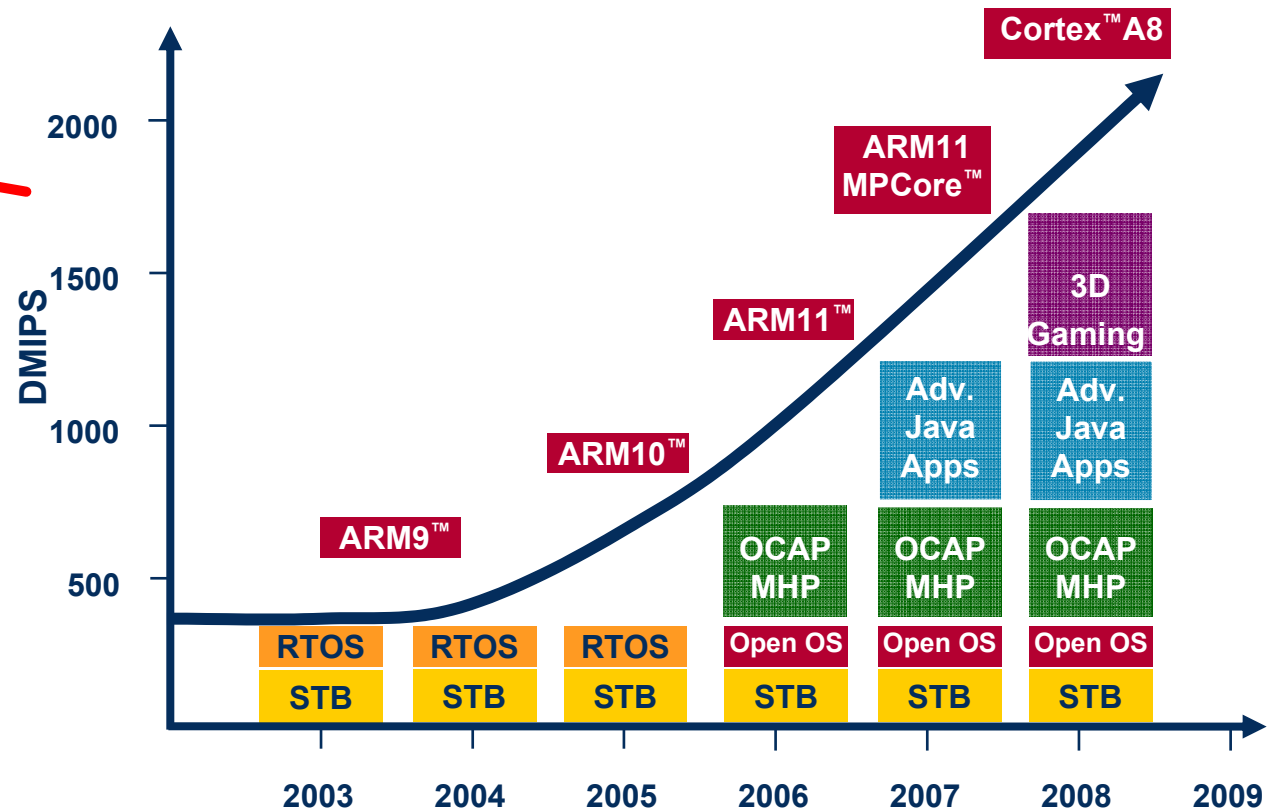
For new consumer device:

High Performance

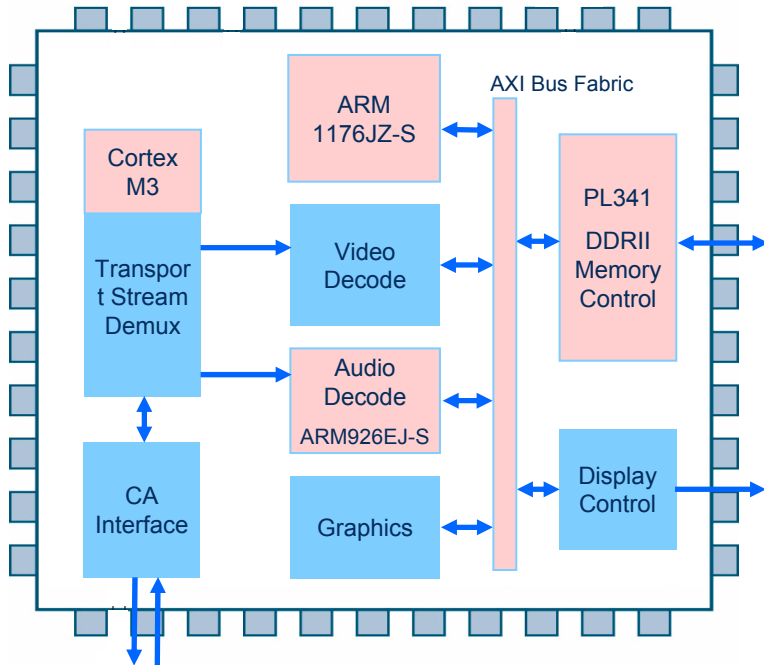
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Next Generation of Product in the Home

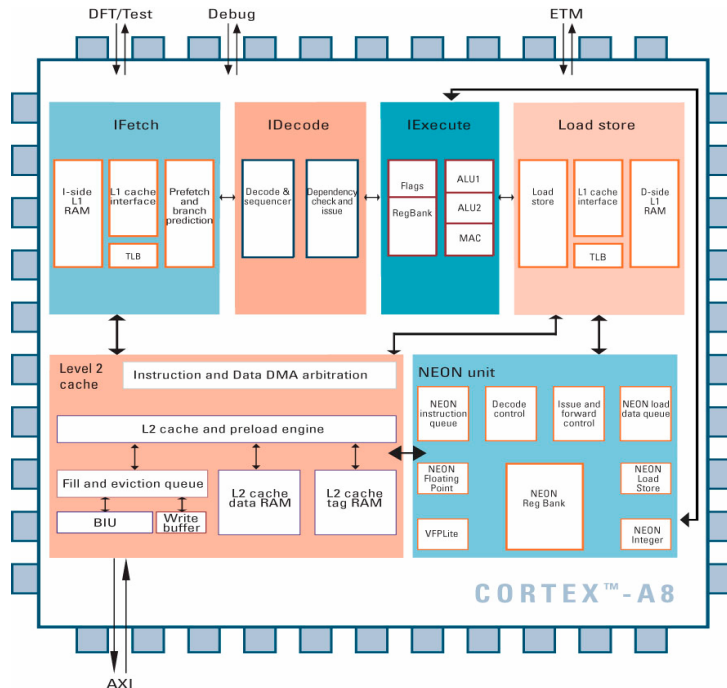


- ARMv6 architecture
- ARM1176JZ-S™ Processor
- 750MHz
- AMBA® 3 AXI™ Memory System
- Fast interrupt mode
- TrustZone® security
- Java Performance



Samsung Blu-Ray Disc Player

Cortex™-A8 Performance



- ARMv7 architecture
 - NEON™ media acceleration
 - Thumb®-2 technology
- Advanced microarchitecture
 - 13-stage superscalar pipeline
 - Advanced branch prediction
- Optimised implementation
 - 1+ GHz and 2,000+ DMIPS on high-performance technologies
 - 600+ MHz and < 300 mW on low leakage 65LP technology
- Synthesizable implementation
 - Flexible deployment and portability using advanced EDA tool flow

cadence

Process	Library	Frequency
CLN90G	Advantage-CE	1 GHz
CLN65LP	Advantage-HS	600+ MHz

Home: Low Power

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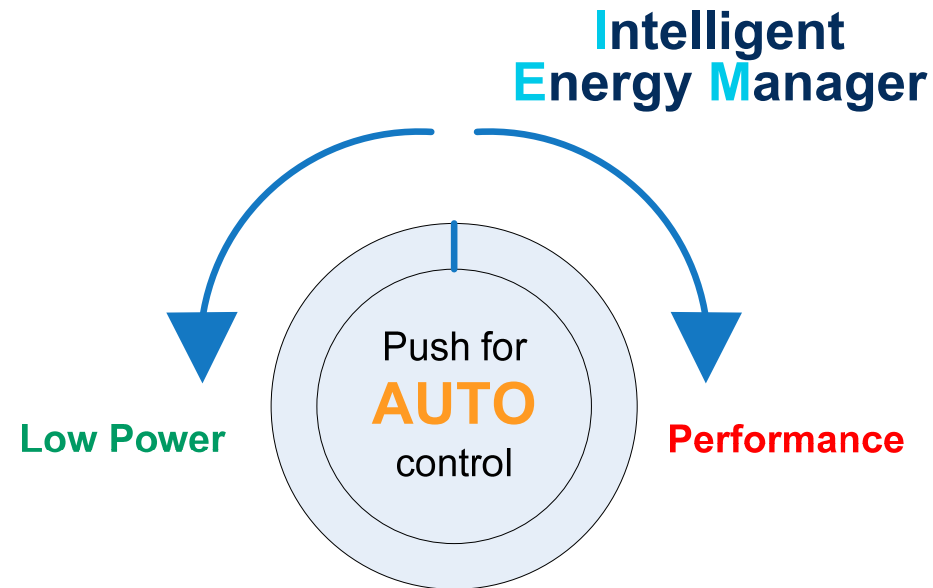
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- Outstanding lower power capabilities are essential for all home devices to reduce manufacturing and running costs and maintain the user experience
- Achieving low power means considering
 - Dynamic run-time control
 - Low power cell libraries
 - CPU design
 - System design
- Balancing power and performance is critical to device success

Dynamic Run-time Control

- More home use cases have the apps. processor active for an increasingly significant percentage of elapsed time
- Standby and idle modes are no longer sufficient to cover these use cases
- Intelligent Energy Manager (IEM) *automatically* adjusts processor performance to software workload
- IEM can be applied to any dynamically scalable SoC design style



$$E = \int_0^t (C(V_{DD})^2 f_c + V_{DD} I_Q) dt$$

The diagram shows the breakdown of the energy equation into two components of power dissipation:

- Dynamic Power Dissipation**: $\int_0^t C(V_{DD})^2 f_c$
- Leakage Power Dissipation**: $\int_0^t V_{DD} I_{lkg}$

The two component boxes are connected by dashed lines to a central box labeled **Total Power Dissipation**.

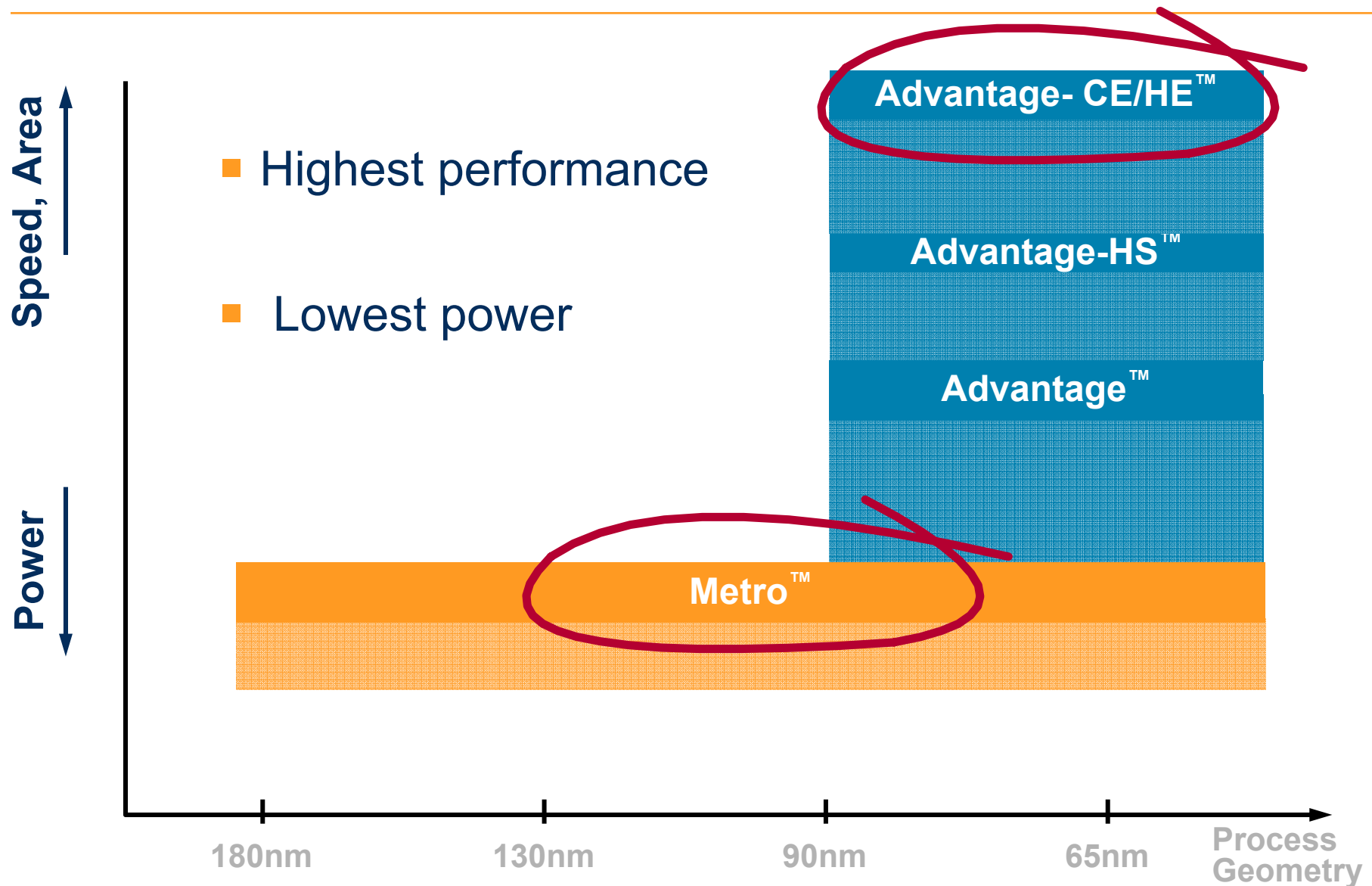
Comprehensive Power Management

- ARM Power Management Kits
- Support for broad range of sophisticated power management techniques
 - Active control of dynamic power
 - Active control of static / leakage power
- Available for all ARM standard cell products
 - Full feature set requires tapless library
- Close collaboration with EDA Partners
 - Reference methodologies
 - Modeling
 - Automation
- Used by the Intelligent Energy Manager



Enable IC designers to achieve 5X reduction in power

Libraries for All Home Devices



Home: Security

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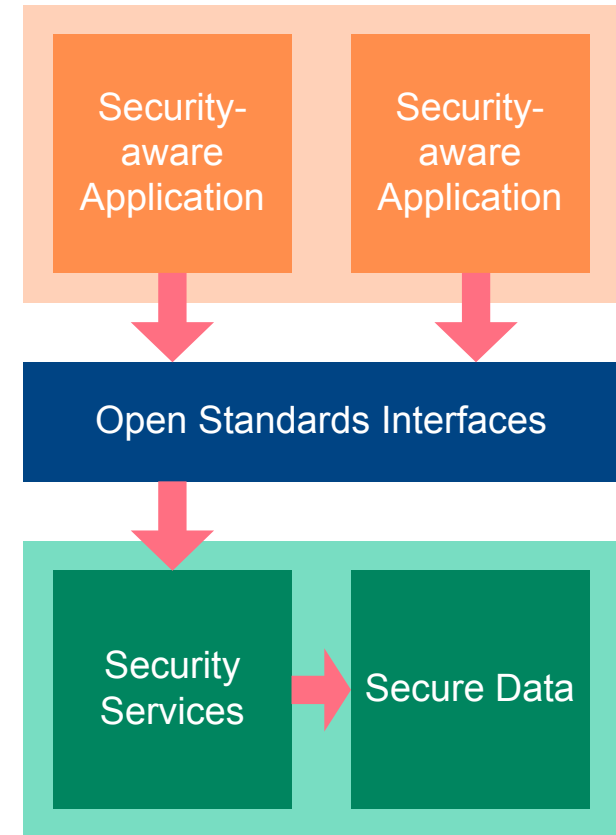
Rapid Integration
Low Risk

- Protecting the value of high quality content is essential, BUT
- Consumers want to be unaware of the operation of content protection schemes
- Consumers want the option to move their content amongst their personal home and mobile devices

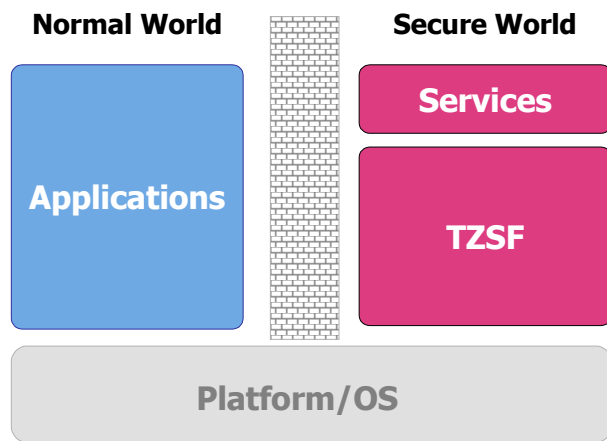
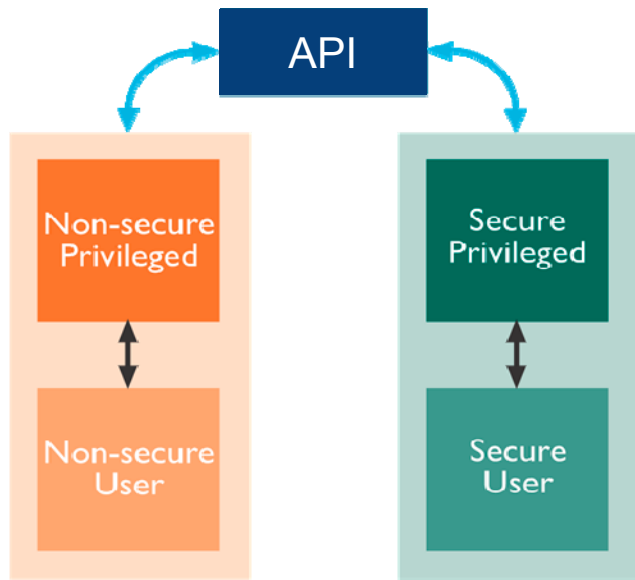
Open and Secure

- Industry standardisation around security is needed to drive faster service adoption
- Standards should be “open” for widest possible adoption
 - “Open” normally means visible and accessible
- But surely secure means “closed”?

TrustZone[®]
Security Foundation by ARM[®]

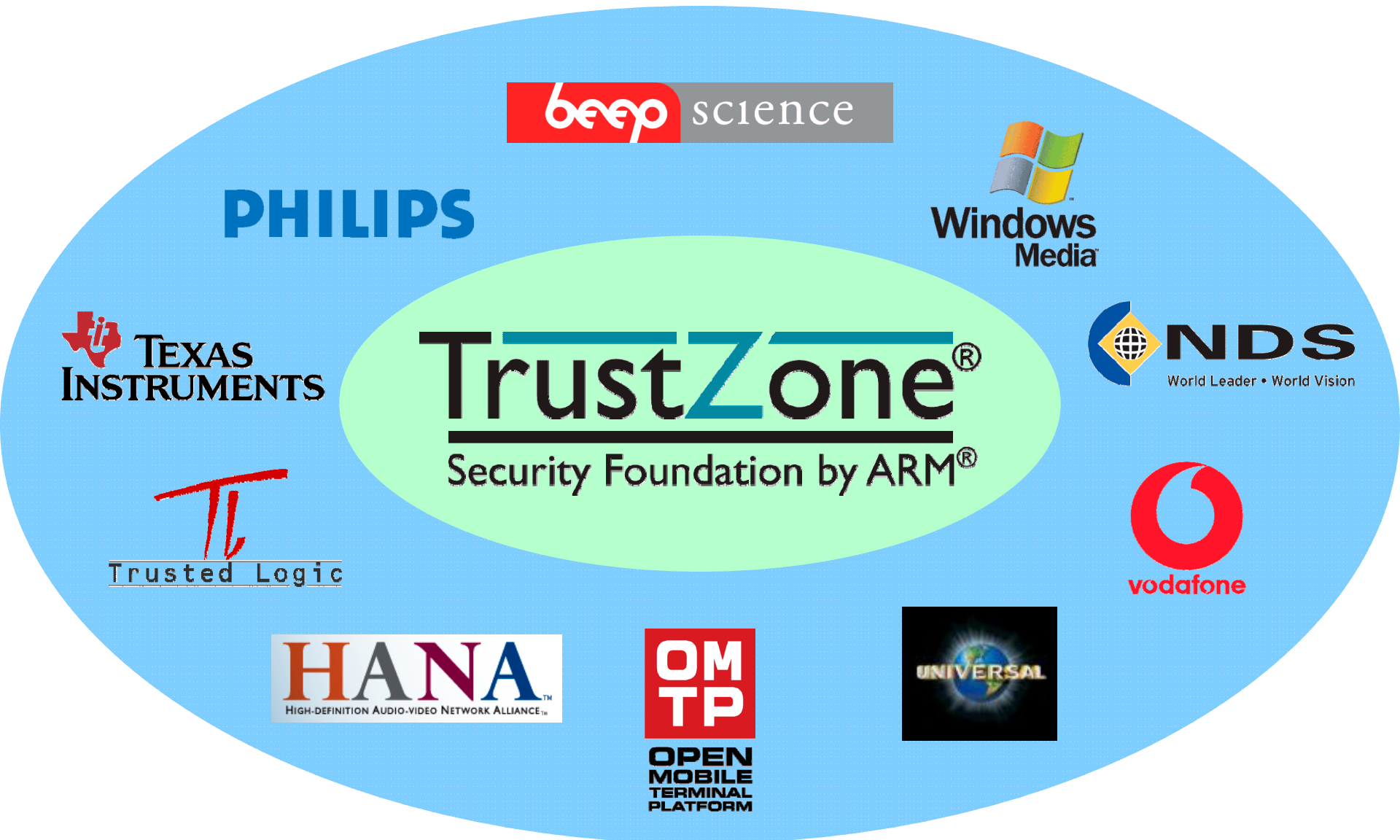


TrustZone is Open *and* Secure



- TrustZone technology adds a “parallel world”
- Normal world for applications
- Secure world for security services
- TrustZone abstracts away the details of the underlying OS and security related hardware
- TrustZone provide a “firewall” between normal and secure worlds
- The TrustZone software API provides access to the security services for authorised applications

TrustZone Ecosystem is Growing



Home: Models

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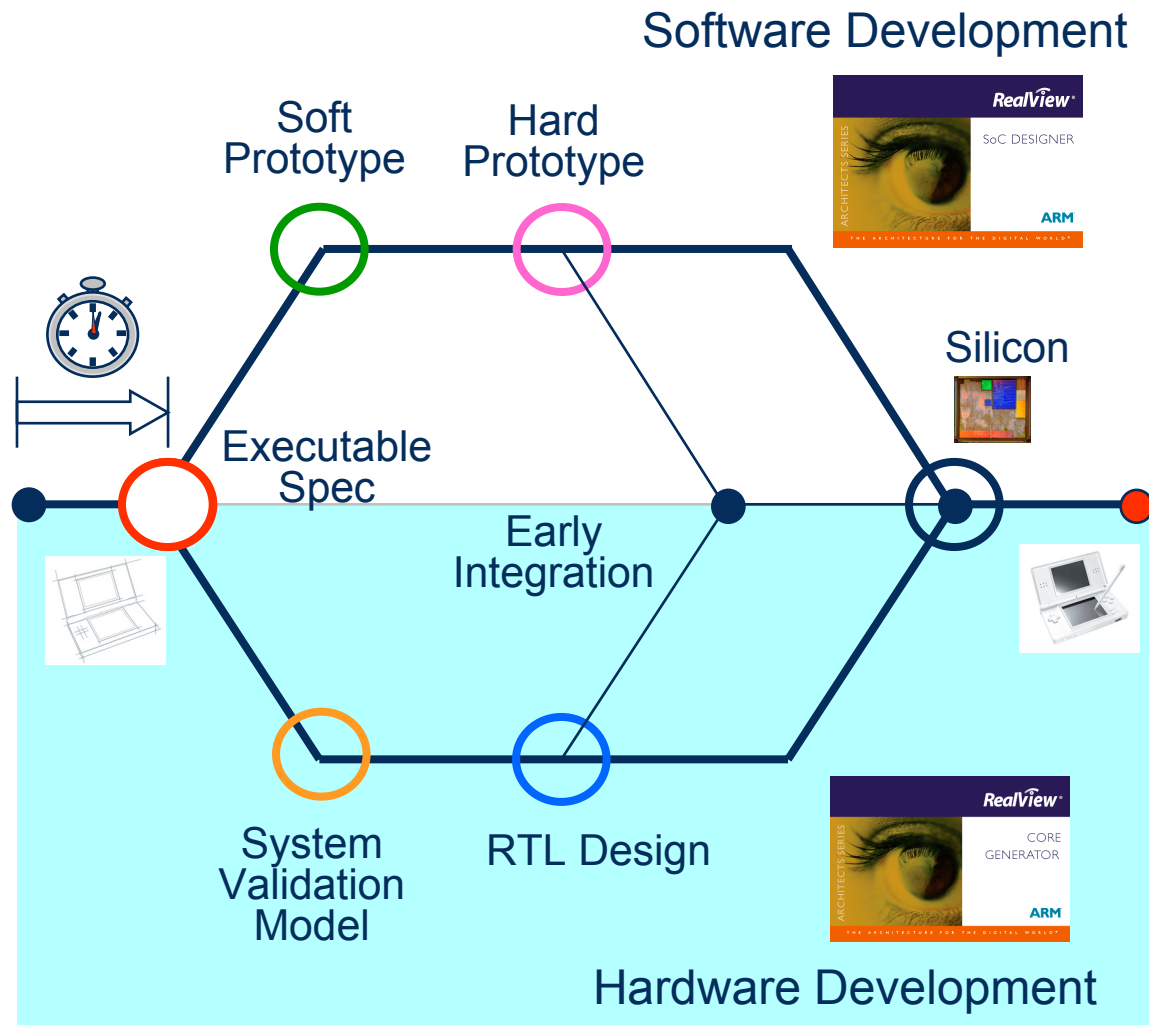
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- Software now accounts more than 50% of the development effort for embedded devices
- Hardware-software co-development is increasingly important in achieving demanding development timescales
- Processor and system models are essential tools to ensure “right first time” delivery
- Both accurate and fast models are needed

Concurrent Development



■ Key theme:

- Keep creating executable targets

■ Target types:

- System exploration model (mixed accuracy)
- System validation model (cycle accurate)
- Soft prototype (fast cycle approx.)
- Hard prototype (hard platform with FPGA config.)
- System test board (silicon)

Home: Ecosystem

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- ARM operates a partnership model
- A healthy, competitive ecosystem provides all Partners with a wide range of choice
 - Silicon design components
 - Fabric components
 - Operating systems
 - Middleware and applications software
 - Development tools
- ARM nurtures the development of multiple ecosystems built on the Connected Community

Diverse Ecosystem

Applications



Security



Middleware



alticast



vidiom systems™

Operating Systems

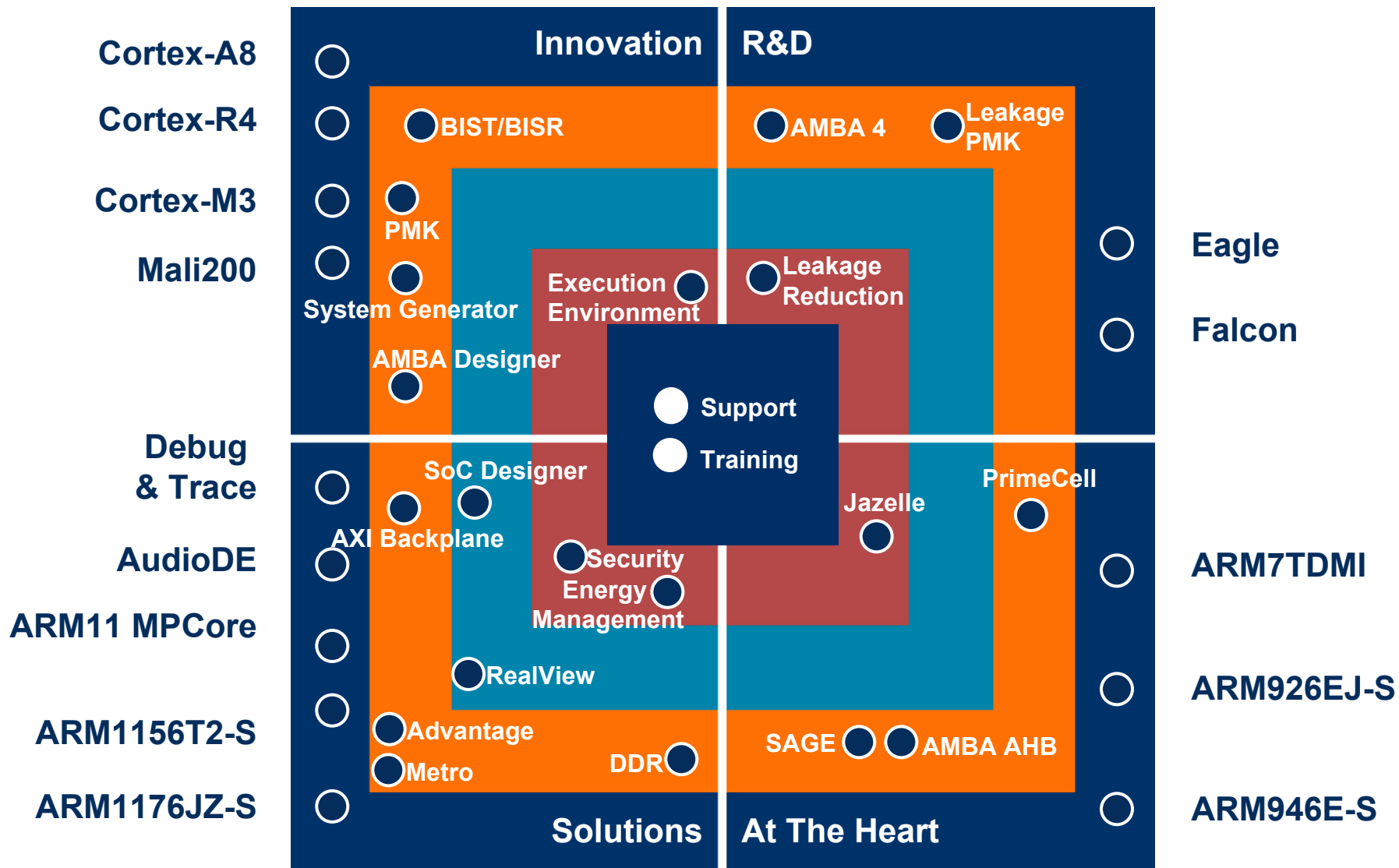


Nucleus RTOS



WIND RIVER

Home



Home: Putting It All Together

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- For the developer of next-generation home devices, ARM delivers the most complete range of technologies available to the market place today and tomorrow
- ARM strives to align these technologies across processor, system and software products and to integrate them with development tools and third party products
- Home: ARM technologies reduce development risk and improve time-to-market